DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

MINIPROJECT REVIEW : 2

SOLAR POWERED WIRELESS CHARGING STATION FOR ELECTRIC VEHICLES (EVS) WITH IOT

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ABSTRACT

The proposed project presents a **Solar Powered Wireless Charging Station for Electric Vehicles (EVs) with IoT.** The system uses solar energy as the primary power source for charging an electric vehicle.

Solar panels convert sunlight into electrical energy, which is regulated and supplied to a wireless charging transmitter. The transmitter coil transfers power wirelessly to a receiver coil placed in the electric vehicle.

An IoT-based monitoring system is used to monitor parameters such as voltage, current, charging status and power consumption. The collected information can be monitored through an IoT platform.

The proposed system provides a **clean, convenient and smart charging solution** while reducing dependence on conventional grid electricity.

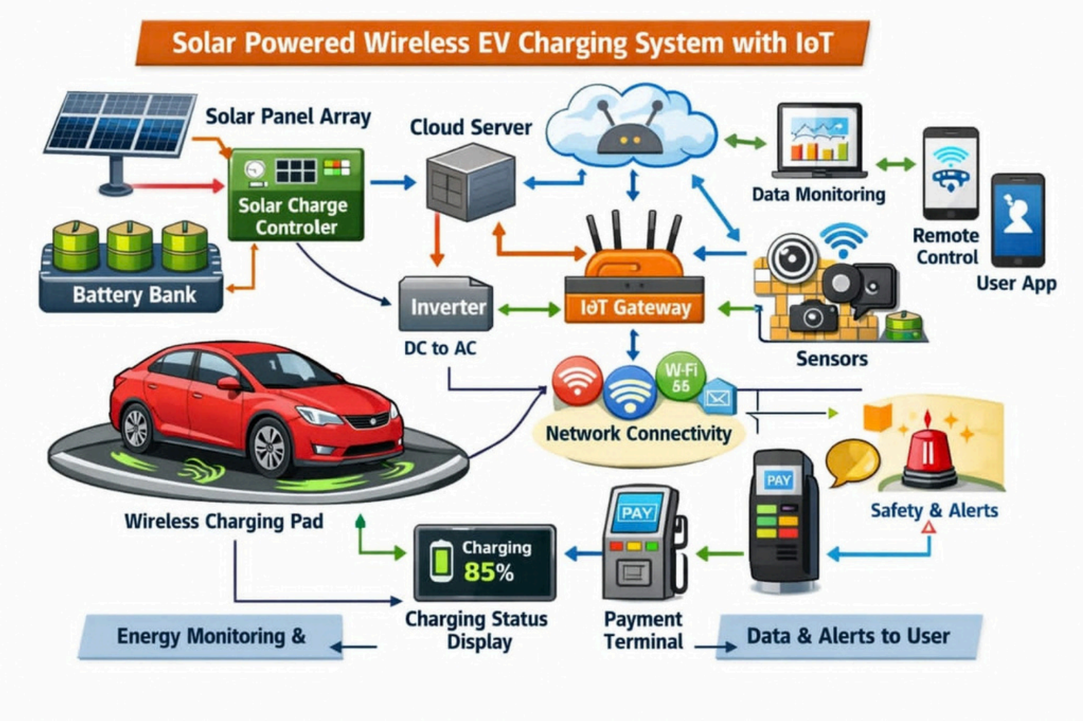
Keywords: Solar Energy, Electric Vehicle, Wireless Charging, IoT, Renewable Energy

INTRODUCTION

Electric Vehicles are becoming increasingly important because they can reduce the use of fossil fuels and harmful emissions from conventional vehicles. However, the availability of convenient and sustainable charging infrastructure is still an important requirement.

Solar energy is a renewable and environmentally friendly energy source. By integrating solar panels with an EV charging station, electricity generated from sunlight can be used for vehicle charging. Wireless charging eliminates the need for physical charging cables. It transfers electrical power using electromagnetic induction between transmitter and receiver coils.

IoT technology can be integrated into the charging station to provide real-time monitoring and smart control



LITERATURE SURVEY

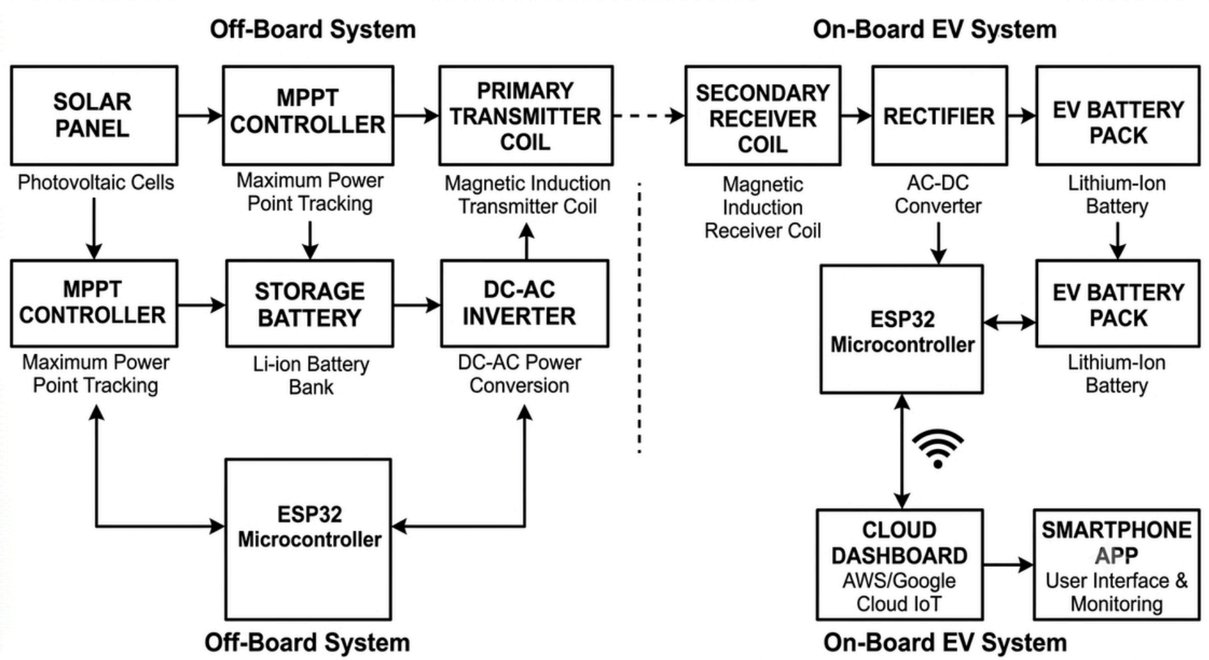
1. Solar-Based EV Charging
Previous studies have investigated the use of photovoltaic panels for supplying energy to EV charging stations. Solar energy reduces dependence on conventional electricity sources.

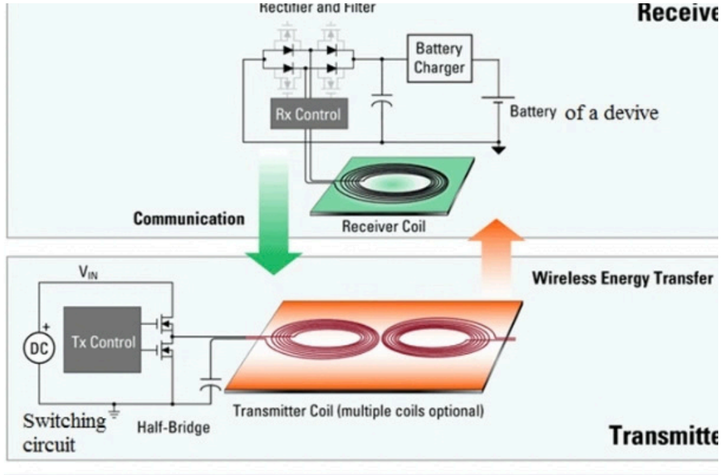
2. Wireless EV Charging
Researchers have developed wireless charging systems using inductive power transfer. These systems allow energy to be transferred without direct electrical contact.

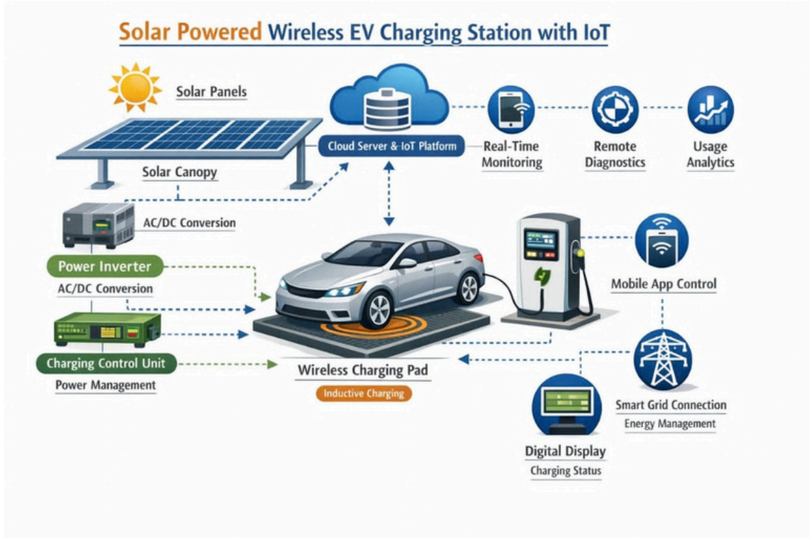
3. IoT-Based EV Monitoring
IoT technologies have been used to monitor voltage, current, battery status and charging conditions remotely.

4. Smart Charging Systems
Smart charging systems combine sensors, controllers and communication technologies to improve charging management and energy utilization

BLOCK DIAGRAM

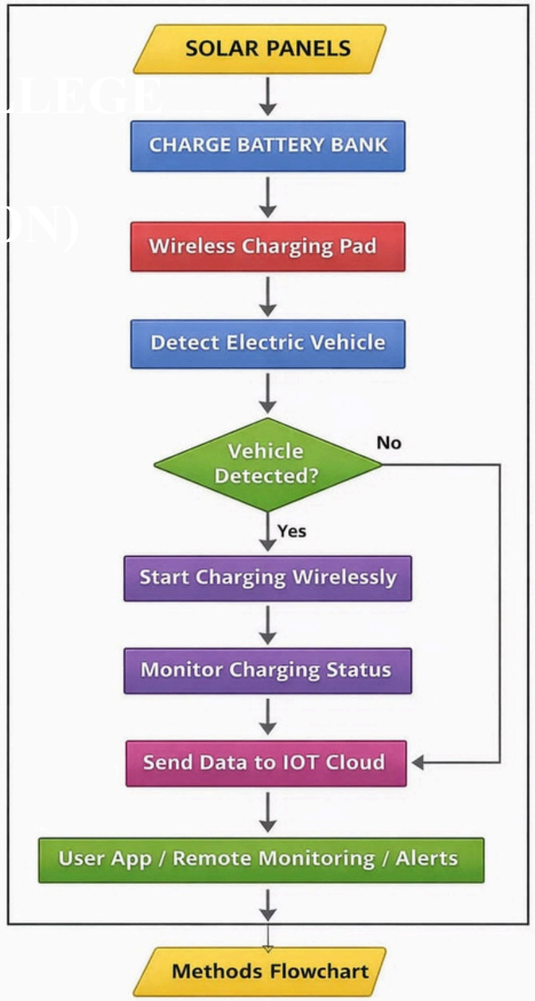




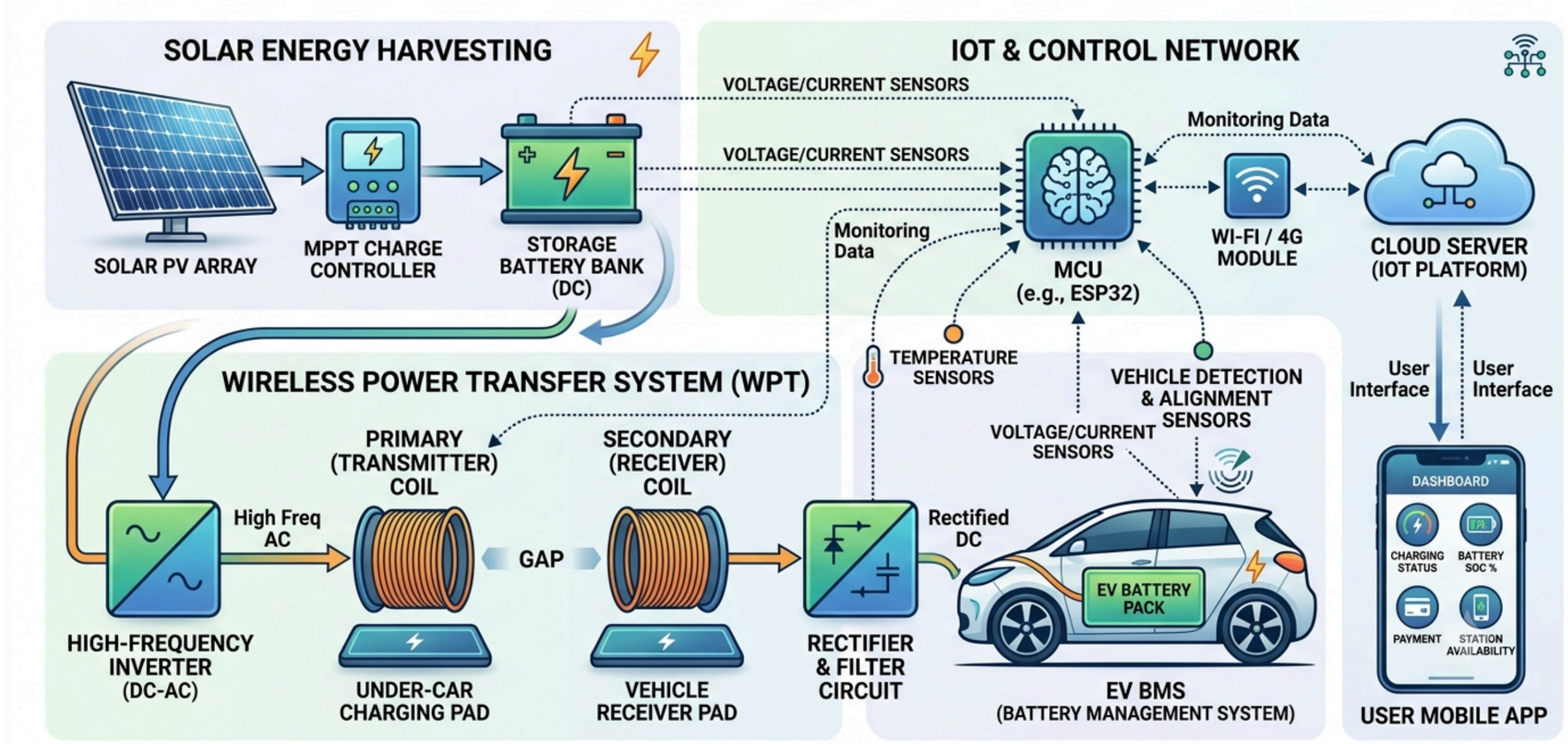


FLOW CHART

1. **Solar panels** – Get energy from sunlight.
2. **Battery** – Stores the solar energy.
3. **Charging pad** – Sends power wirelessly.
4. **Vehicle detection** – Checks if the EV is present.
5. **If vehicle detected** – Charging starts.
6. **Monitor** – Checks the charging status.
7. **IoT Cloud** – Sends charging data online.
8. **User App** – User can see the charging status.

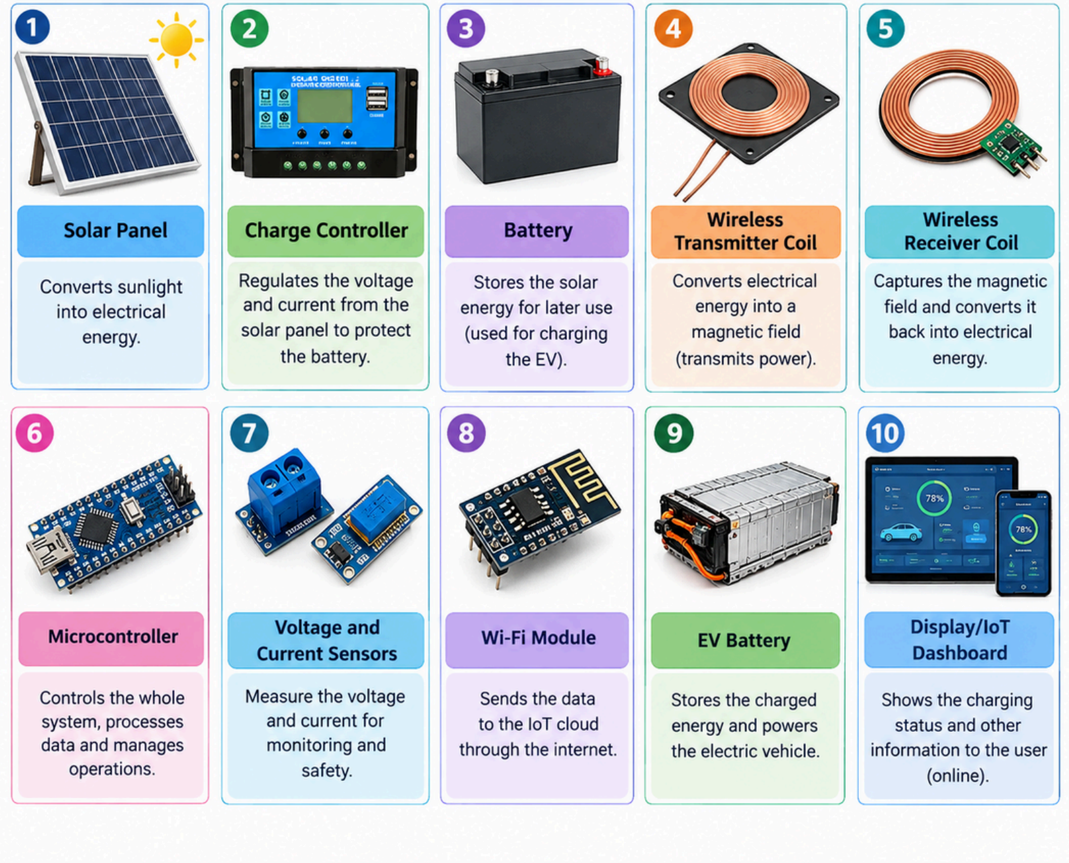


SYSTEM ARCHITECTURE



HARDWARE COMPONENTS DESCRIPTION

1. Solar Panel
2. Charge Controller
3. Battery
4. Wireless Transmitter Coil
5. Wireless Receiver Coil
6. Microcontroller
7. Voltage and Current Sensors
8. Wi-Fi Module
9. EV Battery.
10. Display / IoT Dashboard.



WORKING PRINCIPLE

The system works by combining solar energy, wireless power transfer and IoT monitoring.

- The solar panel receives sunlight and generates DC electrical energy.
- The charge controller regulates the generated energy.
- The battery stores the energy for charging purposes.
- The regulated power is supplied to the wireless transmitter.
- The transmitter coil produces an alternating magnetic field.
- The receiver coil placed in the EV receives energy through electromagnetic induction.
- The received electrical energy is converted and supplied to the EV battery.
- Voltage and current sensors continuously measure charging parameters.
- The microcontroller processes the sensor data.
- The data is transmitted to an IoT platform through Wi-Fi.
- The user can monitor the charging status remotely.
- When the required battery level is reached, charging can be stopped.

SOFTWARE IMPLEMENTATION

- The software section controls the monitoring and communication functions of the system.
- Software Used
- Arduino IDE / Embedded Programming Environment
- C/C++ Programming
- IoT Platform
- Wi-Fi Communication
- Sensor Libraries

RESULTS AND DISCUSSION

The developed system demonstrates the integration of solar energy, wireless EV charging and IoT monitoring.

Observed Results

- Solar energy is successfully converted into electrical energy.
- Wireless power transfer is demonstrated between transmitter and receiver coils.
- EV battery charging can be monitored through electrical parameters.
- Voltage and current values can be measured using sensors.
- Charging information can be transmitted through Wi-Fi.
- The IoT dashboard provides remote monitoring of the charging process.

Result

The prototype demonstrates that a renewable-energy-based wireless EV charging station with IoT monitoring can be developed for smart and sustainable transportation applications.

APPLICATIONS

1. EV Charging Stations
Can be used in dedicated electric vehicle charging areas
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2. Residential Charging
Can be installed in homes with suitable solar power systems.

3. Shopping Malls
Can provide convenient charging facilities in parking areas.

4. Office Parking
Employees can charge EVs while their vehicles are parked.

5. Smart Parking Systems
Wireless charging can be integrated with smart parking infrastructure.

6. Public Transportation
Can support charging infrastructure for electric buses and other electric vehicles.

7. Renewable Energy Charging
Useful in locations where solar energy is readily available.

ADVANTAGES

- Uses renewable solar energy.
- Reduces dependence on conventional grid electricity.
- Provides wireless and convenient charging.
- Reduces the need for physical charging cables.
- Supports IoT-based remote monitoring.
- Provides real-time charging information.
- Can reduce carbon emissions when powered by renewable electricity.
- Suitable for smart charging infrastructure.
- Can be integrated with future smart-city systems.
- Offers potential for automated charging.

CONCLUSION AND FUTURE ENHANCEMENT

- The Solar Powered Wireless Charging Station for Electric Vehicle with IoT combines three important technologies: solar energy, wireless power transfer and IoT.
- The solar panel provides renewable electrical energy, while the wireless charging system transfers power to the electric vehicle without a direct cable connection.
- The IoT system enables monitoring of important charging parameters such as voltage, current, power and charging status.
- The proposed system demonstrates a potential approach for developing convenient, smart and renewable-energy-based EV charging infrastructure. With further improvements in charging efficiency, power capacity, safety and energy management, the system can be developed for larger-scale applications.

SDG/PO/PSO MAPPING WITH JUSTIFICATION

SDG		Goal	Justification
SDG 7		Affordable and Clean Energy	Uses solar energy as a renewable source for EV charging.
SDG 9		Industry, Innovation and Infrastructure	Combines wireless charging, IoT and renewable-energy technologies.
SDG 11		Sustainable Cities and Communities	Supports cleaner transportation and smart charging infrastructure.
SDG 13		Climate Action	Renewable energy-based EV charging can help reduce dependence on fossil-fuel-based energy.

PROPOSED OUTCOME

PO	Mapping Justification
PO1 – Engineering Knowledge	Applies electrical, electronics, IoT and wireless charging concepts.
PO2 – Problem Analysis	Identifies and addresses challenges in EV charging infrastructure.
PO3 – Design/Development	Designs a solar-powered wireless EV charging system.
PO4 – Investigation	Measures and analyzes voltage, current and charging parameters.
PO5 – Modern Tool Usage	Uses microcontrollers, sensors, IoT platforms and wireless technologies.
PO7 – Environment and Sustainability	Uses renewable solar energy for sustainable transportation.
PO12 – Lifelong Learning	Involves emerging technologies such as EV wireless charging and IoT.

PROPOSED OUTCOME

